Customer Sales Analysis

1. Project Overview

This project analyzes customer purchasing behavior using Python and the Pandas data analysis library. The objective of this project is to identify top customers, analyze product sales performance, observe monthly sales trends, and extract useful business insights from sales data. Data analysis and visualization techniques were used to understand patterns in customer transactions and support data-driven decision-making.

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2. Dataset Description

Two datasets were used for this project.

1. sales_data.csv

This dataset contains transaction-level sales information. The dataset includes the following fields:

- Date – Date of the transaction
- Product – Product purchased by the customer
- Quantity – Number of units purchased
- Price – Price per unit
- Customer_ID – Unique identifier for each customer
- Region – Region where the purchase was made
- Total_Sales – Total value of the transaction

This dataset helps analyze revenue trends, product performance, and customer purchasing behavior.

2. customer_churn.csv

This dataset contains information about customer characteristics and subscription details.

Key fields include:

- CustomerID – Unique identifier for customers
- Tenure – Number of months the customer has stayed with the service
- MonthlyCharges – Monthly subscription cost
- TotalCharges – Total amount paid by the customer
- Contract – Type of contract
- PaymentMethod – Payment method used

- PaperlessBilling – Whether the customer uses paperless billing
- SeniorCitizen – Whether the customer is a senior citizen
- Churn – Whether the customer has left the service

This dataset provides insights into customer demographics and behavior.

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3. Data Processing and Analysis Steps

The analysis was conducted using Python with the Pandas, Matplotlib, and Seaborn libraries.

Data Loading

The datasets were loaded into Python using the Pandas `read_csv()` function.

Data Exploration

Initial exploration was performed using functions such as:

- `head()` to preview the data
- `info()` to understand data types and missing values
- `describe()` to view statistical summaries

Dataset Merging

The sales dataset and customer dataset were merged using customer identifiers. This allowed the project to combine transactional data with customer information for deeper analysis.

Data Aggregation

Groupby operations were used to calculate total sales per customer, product, and month.

Visualization

Charts were created using Matplotlib and Seaborn to better understand sales patterns and trends.

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4. Data Visualizations

Several visualizations were created to support the analysis.

Top Customers by Revenue

A bar chart was generated to identify the top customers contributing the highest revenue. This helps businesses recognize high-value customers and focus retention strategies on them.

Monthly Sales Trend

A line chart was created to analyze how sales vary over time. Monthly sales trends help businesses understand seasonal demand and plan inventory accordingly.

Product Sales Analysis

A bar chart was used to analyze total sales by product. This helps identify best-selling products and products that may require marketing improvement.

Sales Heatmap

A heatmap visualization was created to show sales distribution by region and product category. Heatmaps provide a quick visual representation of sales intensity across categories.

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5. Key Insights

The analysis revealed several important insights:

- A small group of customers contributes significantly to total revenue.
- Certain products generate higher sales compared to others.
- Monthly sales trends indicate patterns that may be influenced by seasonal demand.
- Regional variations in sales highlight potential opportunities for targeted marketing strategies.
- Visualizations help simplify complex data and support better decision-making.

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6. Conclusion

This project demonstrates how Python and data analysis libraries such as Pandas can be used to analyze real-world business datasets. By combining customer information with transaction data, meaningful insights were extracted about customer behavior, product performance, and revenue trends.

The analysis shows how data-driven approaches can help businesses identify key customers, optimize product offerings, and improve overall sales strategies.